

Example: Dissimilarity between Asymmetric Binary Variables

Name	Gender	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	M	Y	N	P	N	N	N
Mary	F	Y	N	P	N	P	N
Jim	M	Y	P	N	N	N	N

- Gender is a symmetric attribute (not counted in)
- The remaining attributes are asymmetric binary
- Let the values Y and P be 1, and the value N be 0
- Distance: $d(i, j) = \frac{r + s}{q + r + s}$

$$d(jack, mary) = \frac{0 + 1}{2 + 0 + 1} = 0.33$$

$$d(jack, jim) = \frac{1 + 1}{1 + 1 + 1} = 0.67$$

$$d(jim, mary) = \frac{1 + 2}{1 + 1 + 2} = 0.75$$

		Mary		
		1	0	Σ_{row}
Jack	1	2	0	2
	0	1	3	4
	Σ_{col}	3	3	6

		Jim		
		1	0	Σ_{row}
Jack	1	1	1	2
	0	1	3	4
	Σ_{col}	2	4	6

		Mary		
		1	0	Σ_{row}
Jim	1	1	1	2
	0	2	2	4
	Σ_{col}	3	3	6